

NICKSON ONSOMBI NYABOGA

Software Developer | Backend & Full-Stack (Python, Django, React, Next.js) | Systems & Data

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PROFESSIONAL SUMMARY

Software developer specialising in Python backend engineering and full-stack web development, with experience delivering secure REST APIs, database-driven business systems and modern React and Next.js front ends. Backend engineer on Miti App, an AI-powered agritech platform, where I built the Django REST Framework API with JWT authentication and MySQL, and freelance full-stack developer and ICT officer for a Somaliland-based research and development consultancy. Has integrated M-Pesa payment APIs into billing and point-of-sale systems, and brings strengths in requirements documentation, testing, code review and clear reporting for technical and non-technical stakeholders. About four years of combined academic, internship, freelance and project experience in technology; pursuing a BSc in Computer Science at the Open University of Kenya and holds a Diploma in ICT (final module awarded Distinction).

CORE SKILLS

Programming: Python, TypeScript, JavaScript, SQL, HTML5, CSS3

Backend & APIs: Django, Django REST Framework, RESTful API design, JWT authentication and access control, Django ORM, third-party API integration (M-Pesa Daraja STK Push, SendGrid, AI detection services)

Frontend: React, Next.js (App Router), Tailwind CSS, shadcn/ui, Bootstrap, responsive and accessible (WCAG) interfaces

Databases & Data: MySQL, relational database design, data modelling, data entry and validation, record audits, reports modules, Excel registers and reporting

Engineering Practices: Git and GitHub, code review, debugging and root-cause analysis, Playwright automated testing, CI pipelines, Docker, technical documentation

Systems & Support: Requirements gathering and specifications, change management, stakeholder sign-off, user and application support, software installation and configuration

Tools: Microsoft 365 (Excel, Word, PowerPoint, Outlook, Teams, SharePoint), Google Workspace, Jira, Asana, Slack

IoT: ESP32, environmental sensors, MQTT cloud messaging

PROFESSIONAL EXPERIENCE

Backend Engineer — Miti App (Miti Ni Poa)

MitiNiPoa Africa Group Ltd | AI-powered tree planting and environmental monitoring platform

- Built the platform's REST API with Django, Django REST Framework and MySQL, covering user management, JWT authentication, tree-planting records, AI detection and reports.
- Contribute to the organisation's backend repository, extending the platform with AI services, analytics and notifications.
- Leading remediation of findings from a principal-level backend audit (September 2026) alongside new features, and containerising the service with Docker for consistent environments.
- Built firmware for the platform's ESP32 sensor node, streaming temperature, humidity and soil-moisture data to the cloud over MQTT; resolved GPIO, ADC2/Wi-Fi, calibration and power issues.

Full-Stack Software Developer & ICT Officer (Freelance, Remote)

Siraj Research & Development Consultancy (SRDC) | Hargeisa, Somaliland

- Building the company's multilingual website on Next.js 16, React 19, strict TypeScript and Tailwind CSS v4, with a design-token system, component library and internationalisation for English and Somali, extensible to four more languages.
- Gathered requirements and wrote an 18-document architecture and specification suite before build; manage changes through versioned documentation and phased stakeholder sign-off.
- Set up CI, lint/type checks and Playwright tests; caught defects in review, including a form bug that would have broken submissions and a colour token failing WCAG AA contrast.
- Produce data-driven technical reports, Excel registers, branded templates and presentations for government and development-sector clients, including work under the World Bank-funded Somaliland Food Systems Resilience Project.

ICT Attachee (two industrial attachments)

Kisii County Government | Sep-Dec 2023; Sep-Dec 2024

- Participated in system development and maintenance activities within the county ICT department.

- Installed, upgraded and configured software, provided first-line application support to staff, and performed hardware diagnostics.
- Handled data entry, documentation and records preparation.

SELECTED PROJECTS

Point-of-Sale System with M-Pesa

Django, MySQL, Daraja API

- Built a POS application with M-Pesa STK Push payments, tested against the Safaricom Daraja sandbox.

WiFi Billing System

Django, MySQL, Daraja API

- Built a billing platform with package management, user authentication and secure online M-Pesa payments.

Online Shoe Store

Web application

- Built an e-commerce platform with product listings, cart, user authentication, order management and a responsive UI.

Membership & Library Management Systems

Python, Django 4.2

- Built record-management systems covering members, stock, cataloguing, loans and reporting.

Client Websites

HTML, CSS, JavaScript

- Developed websites for Nyanchwa Adventist Primary School, Amatech Systems and Hon. Patrick Lumumba.

EDUCATION

Bachelor of Science in Computer Science (in progress)

Ongoing

Open University of Kenya | Fully online, flexible study schedule

Diploma in Information Communication Technology

Sep 2022 – Jul 2025

Kisii National Polytechnic | KNEC | Module III: Distinction; Modules I and II: Credit

- Top grades (1–2, where 1 is highest): Database Management Systems, Internet-Based Programming, Management Information Systems, Computer Applications, Visual Programming, Course Specialisation Project.
- Relevant coursework: Systems Analysis and Design, Object-Oriented Programming, Structured Programming, Quantitative Methods.

Kenya Certificate of Secondary Education (KCSE)

2021

E.L.C.K. Itierio Boys High School

PROFESSIONAL TRAINING

Website Development Boot Camp — Certificate of Completion

August 2025

eMobilis Technology Training Institute | Full scholarship funded by the Mastercard Foundation

- Python and Django web development and full-stack web application development.

REFEREES

Dr Charles Obiero — Project Lead, Miti Ni Poa Innovation Project | Contact details available on request

Zakari Raghe — Managing Director, Siraj Research & Development Company | Contact details available on request

Isaiah Osoro — ICT Director, Kisii County Government | Contact details available on request